

PAPER • OPEN ACCESS

Two- and three-body attachment, electron transport and ionisation in water-air mixtures

To cite this article: J de Urquijo *et al* 2024 *J. Phys. D: Appl. Phys.* **57** 125205

View the [article online](#) for updates and enhancements.

You may also like

- [Electron collision cross section set of O₂ and electron transport coefficients in O₂ and O₂-Ar mixtures](#)
S Kawaguchi, Y Iwabe, K Takahashi et al.
- [Scanning drift tube measurements and kinetic studies of electron transport in CO](#)
S Dujko, D Bošnjakovi, M Vass et al.
- [On the different regimes of gas heating in air plasmas](#)
Carlos D Pintassilgo and Vasco Guerra

Two- and three-body attachment, electron transport and ionisation in water-air mixtures

J de Urquijo^{1,*} , O González-Magaña¹ , E Basurto²  and A M Juárez¹ 

¹ Instituto de Ciencias Físicas, Universidad Nacional Autónoma de México, 62251 Cuernavaca, Morelos, Mexico

² Departamento de Ciencias Básicas, Universidad Autónoma Metropolitana, Av. San Pablo 420, Azcapotzalco, Ciudad de México 02128, Mexico

E-mail: jdu@icf.unam.mx

Received 8 August 2023, revised 25 November 2023

Accepted for publication 15 December 2023

Published 27 December 2023



Abstract

Three-body electron attachment in the mixtures of H₂O and dry air have been measured over a wide range of the density-reduced electric field, E/N , from 3–130 Td and gas pressures, for mixture combinations ranging from 1% to 50% of H₂O. We have measured the regions of three-body attachment (3–30 Td) and two-body dissociative attachment (40–130 Td). Besides, the increasing amount of H₂O in the mixture causes an increase in the three-body reaction rates of up to two orders of magnitude in comparison with that measured for dry air. On the other hand, the three-body attachment coefficients exceed the two-body ones (dissociative attachment) at high pressures. Good agreement has been found with previous measurements of H₂O-dry air mixtures with H₂O concentrations of up to 2%. We know of no previous work for higher H₂O concentrations. Values of the effective ionisation coefficients and longitudinal diffusion coefficients derived from the same measurements are also presented.

Keywords: three-body attachment, water-air, electron transport, dissociative attachment, two-body attachment, pulsed Townsend

1. Introduction

Gaseous discharges and low-temperature plasmas generated in moist air, near a water surface or in bubbles and foam are present in a wide variety of natural phenomena and technological applications. For instance, water vapour in dry air plays a significant role in plasma medicine [1, 2], air–water microwave plasma torches [3], plasma-assisted combustion [4], aerodynamics, purification of gaseous and liquid media, plasma agriculture, streamers [5], and sprites [6]. Among other

effects, humidity increases the breakdown voltage of moist air discharges, mainly due to the increased electron attachment as compared with dry air [7] and is also one of the most important atmospheric parameters which affect corona discharges [8]. In some cases, water vapour in a plasma can be present as a contaminant, producing undesirable effects such as corrosion [9]. Recently, plasma-activated water [10] has gained much attention due to its importance in agriculture [11], purification, and biomedical applications.

Because of its importance in plasma modelling, several databases of cross-sections and/or swarm parameters for the interaction of electrons with water have been produced, as well as those for dry/synthetic air and its mixtures [12–14].

This paper is mostly dedicated to report on the measurement of two- and three-body electron attachment in H₂O-dry air mixtures. Pioneering experimental work in this direction was carried out by Chanin, Phelps and Biondi [15, 16] using

* Author to whom any correspondence should be addressed.

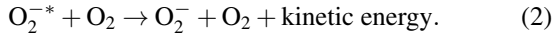


Original content from this work may be used under the terms of the [Creative Commons Attribution 4.0 licence](https://creativecommons.org/licenses/by/4.0/). Any further distribution of this work must maintain attribution to the author(s) and the title of the work, journal citation and DOI.

a pulsed drift tube to determine the attachment coefficients of O_2 . They found that for average electron energies below 1 eV, three-body attachment takes place by a two-step process, the first one leading to the capture of the electron, leaving the oxygen molecule in an excited state:



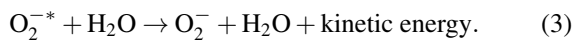
followed by stabilisation via the collision with a second O_2 molecule:



Three-body attachment in O_2 was later confirmed by Spence and Schultz [17] using an experiment with an ion beam setup and by Grünberg [18] who, using a pulsed Townsend experiment was able to measure the three-body attachment coefficients in pure O_2 over a wide range of gas pressures from 14.9 to 880 Torr, and density-reduced electric field strengths, E/N , from 0.3 to 54 Td (1 Townsend = 10^{-17} V cm²). With Grünberg's and Chanin's data, Taniguchi *et al* [19] used a Boltzmann equation method to derive a density-dependent, three-body attachment cross section, and having achieved a fair to good agreement with the above experiments.

Experiments on dry and humid air have been focused mainly on mixtures containing 1%–2% H_2O in dry air, which is the usual abundance of water in atmospheric air [20]. Nevertheless, one may think of other situations in which the water contents in air are higher such as in the flue gases, discharges between clouds, some industrial processes and bioplasma applications.

Three-body reactions with a chaperon molecule other than O_2 have been measured by Chanin *et al* [16], using He and N_2 , for which the first step is the same as that given in Reaction (1), followed by collisional stabilisation with an H_2O molecule, which is the present case:



Even though H_2O contains an oxygen atom, it is unlikely that at the low electron energies corresponding to the E/N range 1–36 Td, H_2O may undergo dissociation. H^- , O^- and OH^- are produced by electron impact on H_2O [14]. According to the electron attachment cross section set revised by Itikawa and Mason [14], they recommend the electron impact dissociative cross sections measurements of Melton [21], and Compton and Christophorou [22]. It turns out that the H^- yield is nearly 30 times higher than that of O^- and OH^- at their first energy peaks, occurring between 6.32 and 6.44 eV with cross sections of 6.37×10^{-18} cm², 0.133×10^{-18} cm² and 0.116×10^{-18} cm², respectively.

Using Bolsig+ [23], a calculation of the electron energy distribution functions (EEDF) was performed for the H_2O -dry air mixtures over the E/N range 1–30 Td. The EEDF curves for $E/N = 1$ Td (broken lines) and $E/N = 30$ Td (solid lines) are shown in figure 1, together with a vertical line indicating the first energy peaks for H^- , O^- and OH^- . Thus, one finds that for $E/N = 30$ Td and at an electron energy of 6.3 eV,

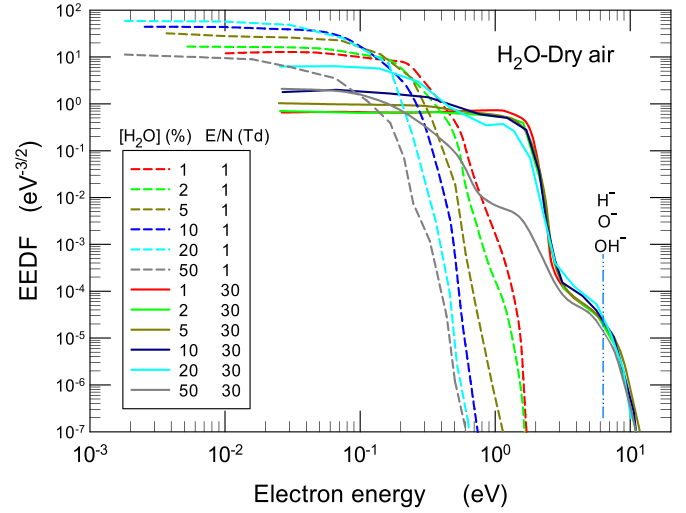


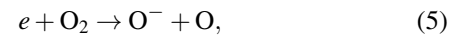
Figure 1. The electron distribution functions, EEDF, calculated from Bolsig+ [23] for the 1%–50% H_2O -dry air mixtures, and E/N values over the range 1–30 Td. The vertical line around 6.3 eV indicates the peak electron energy values for the formation of H^- , O^- and OH^- by dissociative electron attachment (See the text).

the EEDFs are about four orders of magnitude smaller than those prevailing at lower energies in the range 30–100 meV. Therefore, in what follows, we can safely assume that the only source of O_2^- in the avalanche over the E/N range 1–30 Td comes from the O_2 in dry air.

When the electron energy exceeds the above 6.32–6.44 eV threshold, water dissociates into the above three fragments, which in turn means that a two-body or binary collision process is operative. This in turn means that the density-normalised attachment coefficient, η/N , would not depend on the gas density, N , and the reaction is



where X^- stands for H^- , O^- or OH^- . Processes (1)–(4) are the core of this work and will be discussed at length in section 3. Likewise, when the energy of electrons exceeds the dissociation energy of O_2 , O^- is formed by the dissociative attachment process



with cross sections peaking at 5.61 eV and 7.89 eV, leading to O^- in the 2P and 1S states, respectively [24]. In the context of our experiment, we shall show that the latter process takes place for $E/N > 40$ Td.

2. Experimental apparatus and analytical procedure

The pulsed Townsend apparatus was used for the present measurements and has been described in detail previously [25]. Briefly, it consists of a parallel-plate capacitor in the form two discs with 12 cm in diameter and round edges. The initial photoelectrons are produced by the incidence of a UV

laser-light pulse with 3–5 ns duration, 10 Hz repetition frequency and up to 1 mJ pulse energy from the third harmonic (355 nm) from a Nd-YAG laser. The total displacement current pulse due to the motion of charge carriers (electrons and ions) across the electrode gap is measured by a 40 MHz, 10^5 V A^{-1} transimpedance amplifier connected to the cathode. The anode has a 2 cm diameter copper mesh (2.8 lines per mm crosswise) to allow the passage of the laser light beam while maintaining the electric field homogeneity. The anode-cathode distance was accurately fixed at either 30 mm or 31 mm to an accuracy of 0.025 mm. The gas pressure inside the discharge chamber was continuously monitored by a 1000 Torr absolute pressure capacitance gauge to an accuracy of 0.025%. The high voltage applied to the cathode was provided by a regulated power supply. The electron avalanche pulses were registered by a 100 MHz bandwidth oscilloscope and stored in a computer for further analysis. All the measurements herein presented were performed at room temperature (293–300 K). The overall gas pressure range was 1–307 Torr. The high pressures were used to achieve low E/N values, thereby enabling the observation of the pressure dependence of the attachment coefficient. The effective ionisation and longitudinal diffusion coefficients and the electron drift velocities were measured over a lower pressure range between 1 and 15 Torr.

The H_2O -dry air mixtures were prepared inside the chamber. Liquid water of gas chromatographic grade (Sigma Aldrich) was used. The maximum water vapour pressure allowed in the experiments was 16 Torr, which is 25% lower than the saturation vapour pressure at 298 K (25 °C). Dry air with a purity of 99.8% was flushed directly from the cylinder into the vacuum chamber.

The flux drift velocity W_F , the flux density-normalised longitudinal diffusion coefficient $ND_{L,F}$ and the effective ionisation coefficient $\alpha_e/N = (\alpha - \eta)/N$, respectively, were derived from an analysis of the electron current pulses using the equation derived in [26] under the assumption that all photoelectrons are released as a Dirac-delta pulse from the cathode at $t = 0$.

$$I_e(t) = \frac{n_0 q_0 W_F}{2d} \exp(\alpha_e W_F t) \left\{ \left[1 - \Phi \left(\frac{(W_F + \alpha_e D_{L,F})t - d}{\sqrt{4D_{L,F}t}} \right) \right] + \exp \left(\frac{W_F + \alpha_e D_{L,F}}{D_{L,F}} d \right) \left[\Phi \left(\frac{(W_F + \alpha_e D_{L,F})t + d}{\sqrt{4D_{L,F}t}} \right) - 1 \right] \right\} \quad (6)$$

where n_0 , q_0 and d are the initial number of photoelectrons, electron charge and gap distance, respectively; $\alpha_e = \alpha - \eta$ is the effective ionisation coefficient, where α and η are the electron impact ionisation and attachment coefficients, respectively. $\Phi(x)$ is the error function of argument x . The argument of the first error function in equation (6) is negative for

$$t < \frac{d}{W_F + \alpha_e D_{L,F}}. \quad (7)$$

Then, as $D_{L,F} \rightarrow 0$, the value of the whole term in brackets is $1 - \Phi(-\infty) = 2$. On the other hand, the argument of the second error function is always positive; hence, as

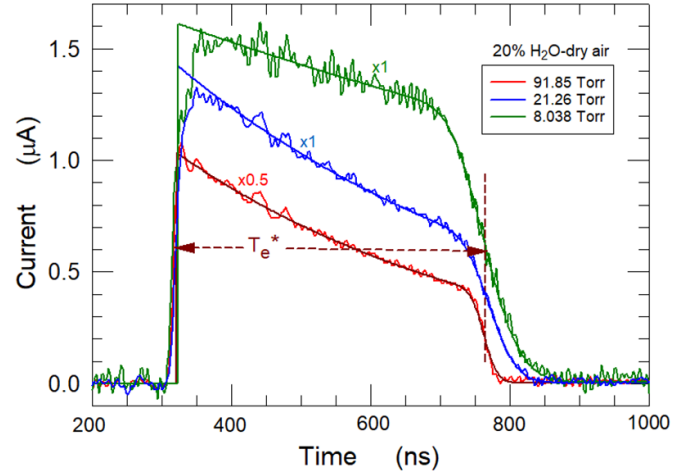


Figure 2. Sample pulses in 20% H_2O -dry air, $E/N = 50 \text{ Td}$, at three different pressures (see inset) and their fitting curves calculated according to equation (6). Since $D_{F,L} \sim 1/N$, note that the diffusion tail becomes more pronounced as the total gas pressure (density) is reduced. The vertical bar at $\sim 760 \text{ ns}$ indicates the approximate position of the half-height time on the falling edge of the pulse, and T_e^* is the approximate value of the electron transit time (see text). The fast rise of the calculated curves from equation (6) is due to the assumption that the initial photoelectrons are liberated from the cathode as a Dirac-delta pulse.

$D_{L,F} \rightarrow 0$, the whole term in brackets is $\Phi(\infty) - 1 = 0$. Hence, equation (1) becomes

$$I_e(t) = \frac{n_0 q_0 W_F}{d} e^{\alpha_e W_F t}, \quad (8)$$

which is valid for $0 \leq t \leq d/W_F$. This equation has been derived in [27, 28]. Moreover, for any value of $D_{L,F}$ when $t = 0$, equation (6) becomes

$$I_e(0) = \frac{n_0 q_0 W_F}{d}. \quad (9)$$

The initial photoelectron number n_0 , can be calculated by firstly making an estimate of the electron drift velocity

$$W_F^* = d/T_e^*,$$

where T_e^* is the mean electron transit time. Referring to figure 2, where three sample transients at the same E/N and different total mixture pressures are shown, let us choose the one for which $p = 8.038 \text{ Torr}$. The starting time of the avalanche is that at which its rising edge reaches half its maximum or, in other words, a value corresponding to the maximum of the Gaussian-like laser pulse signal used. Thus, with $I_e(0) = 1.06 \mu\text{A}$, $d = 3.1 \text{ cm}$, $q_0 = 1.6 \times 10^{-19} \text{ C}$ and $T_e^* = 447 \text{ ns}$, we get $n_0 = 3 \times 10^6$ initial photoelectrons. Knowledge of n_0 is useful to ensure that the local field produced by the moving space charge is negligible in comparison with the external field, E . Typically, values of n_0 in our experiment are in the range 10^5 – 10^7 electrons. On the other hand, n_0 is not needed to fit the measured avalanche pulses since the term $n_0 q_0 / 2d$ in equation (6) is a constant.

In figure 2, to within 1% uncertainty and regardless of the pressure, the three electron transit times coincide very approximately. Denoting the initial, trial value α_e , as α_e^* , and taking the logarithm of equation (12), we obtain

$$\ln[I_e(t)] = \ln\left(\frac{n_0 q_0 W_F^*}{2d}\right) + \alpha_e^* W_F^* t, \quad (10)$$

which is the equation of a straight line of slope $m = \alpha_e^* W_F^*$. Since W_F^* is already known, then α_e^* can be calculated. Entering these two initial parameters into the computer programme to fit the observed pulses according to equation (6), and after small adjustments to α_e^* and W_F^* to obtain the best fit, final values of W_F , $\alpha_e = (\alpha - \eta)/N$ and $ND_{L,F}$ are determined.

3. Results

Most of the values reported in this paper were the result of at least three measurements of the swarm coefficient at different pressures. Thus, the error bar associated to them corresponds to the average value. An exception to the above are the measurements of the H₂O-dry air mixtures for $E/N < 30$ Td, where a clear pressure dependence was found. The overall uncertainties for the measured swarm parameters are given in table 1 below and the values for W_F , $(\alpha - \eta)/N$, ND_L and η/N^2 are tabulated as a function of E/N in the appendix (see tables 2 to 5). The rate constants can be evaluated from these values and their uncertainties.

3.1. Electron drift velocities

The electron flux drift velocity, W_F , in H₂O, dry air and their mixtures is plotted in figure 3 as a function of E/N . These velocities have already been reported [29] and are repeated here, excepting those for the 50% H₂O-dry air mixture, which have been measured recently. The overall uncertainty in this swarm parameter is 1%–1.5%. It must be noted that over the whole range of measurement, the electron drift velocities were found independent on gas density (pressure).

3.2. Longitudinal diffusion coefficients

The density-normalised longitudinal diffusion coefficients of electrons in H₂O, dry air and their mixtures are plotted in figure 4 as a function of E/N . It is interesting to note the increasing influence of H₂O on this coefficient for H₂O contents higher than 10%.

3.3. Effective ionisation and two- and three-body attachment coefficients

3.3.1. Two-body attachment in H₂O-dry air mixtures. The effective ionisation coefficients, $(\alpha - \eta)/N$, where α and η are the ionisation and attachment coefficients, respectively, are plotted in figure 5 over the E/N range 40 Td–400 Td, displaying the regions for which two-body, dissociative attachment

Table 1. Uncertainties in the measured swarm coefficients (%).

	W_F	$ND_{L,F}$	$(\alpha - \eta)/N$	$k_{\text{ion eff}}$	η/N^2	$k_{3\text{-body}}$
	cm s ⁻¹	cm ² V ⁻¹ s ⁻¹	cm ²	cm ³ s ⁻¹	cm ⁵	cm ⁶ s ⁻¹
1%	1	15	6–8	7–9	15	25
2%	1	15	6–8	7–9	8	9
5%	1	15	6–8	7–9	7–14	8–15
10%	1	15	4–8	5–9	16	17
20%	1	20	5–8	6–9	12	17
50%	1	20	15	16	7–14	8–15
H ₂ O	1.5–2	10	6	8	8	—
Dry air	1.5	15	8	9.5	6–10	—

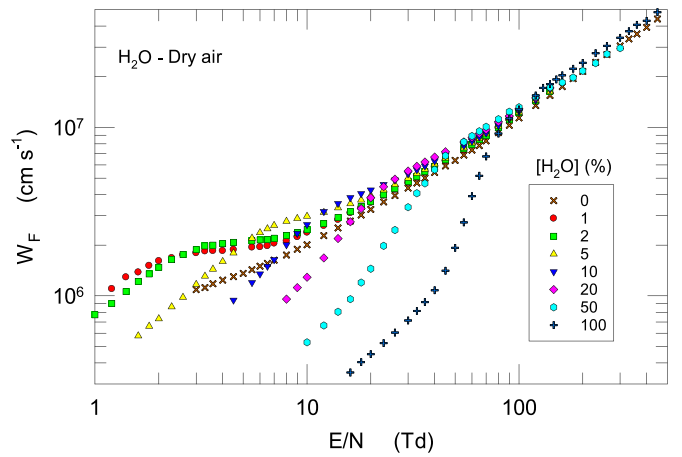


Figure 3. Electron flux drift velocity in H₂O-air mixtures. Except for the 50% H₂O-dry air mixture, these coefficients were taken from [29]. Reproduced from [29]. © IOP Publishing Ltd All rights reserved.

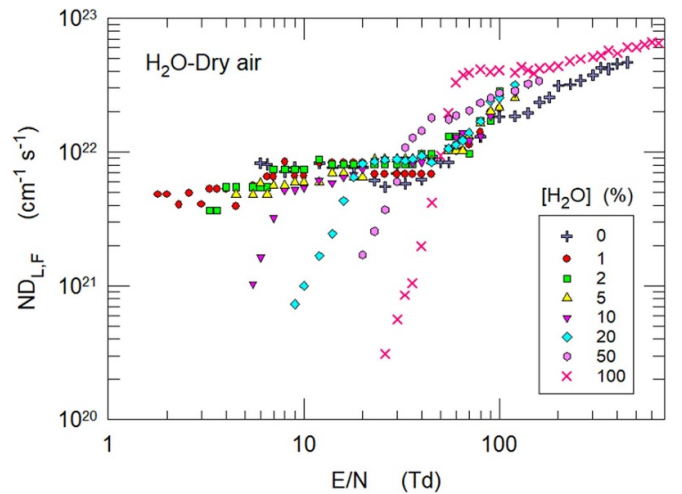


Figure 4. Density-normalised longitudinal diffusion coefficient (flux) for electrons in H₂O-air mixtures.

predominates at the lower E/N range and those of ionisation where $(\alpha - \eta)/N > 0$. For the sake of clarity, the inset contains the curves for which the H₂O share in the mixture ranges between 1% and 20%, whereas the 50% H₂O-dry air, dry air

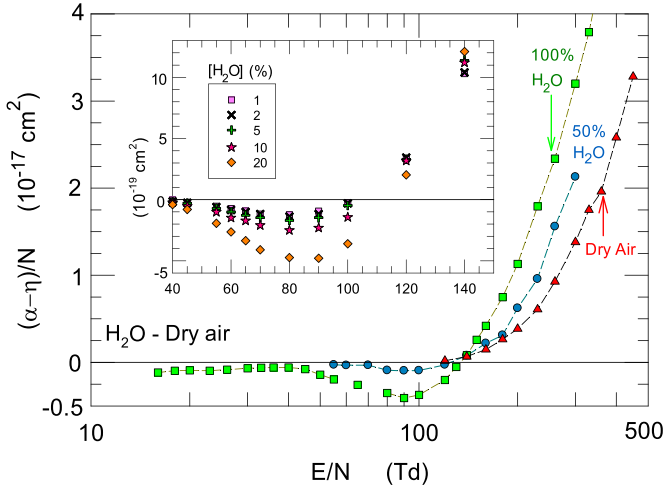


Figure 5. Effective ionisation coefficients for H₂O-air mixtures due to binary collisions only, leading to the formation of O⁻ by electron impact dissociation of O₂.

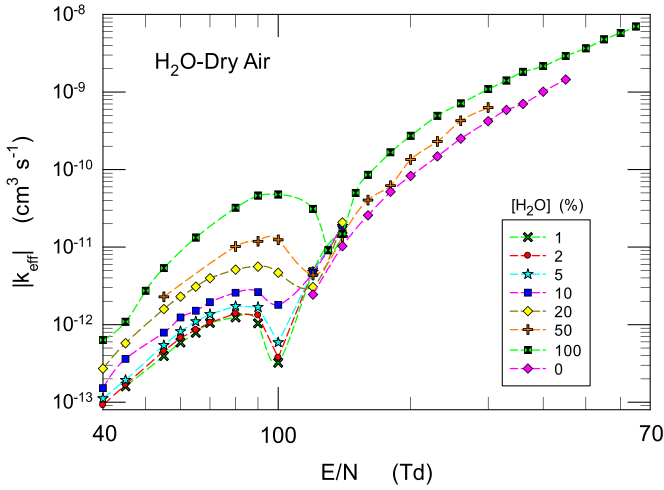


Figure 6. The absolute value of the two-body effective ionisation rates, $|k_{\text{eff}}|$, for the H₂O-Dry Air mixtures. The $(\alpha-\eta)/N$ values to the left of the minimum of each curve are negative.

and 100% H₂O are plotted on the main frame. These curves show clearly how the electronegative character of the mixtures increases with the H₂O concentration.

Knowledge of $(\alpha-\eta)/N$ and W_F allows the calculation of the two-body effective rate of ionisation as

$$k_{\text{eff}} = \frac{\alpha - \eta}{N} W_F. \quad (11)$$

The absolute values of k_{eff} are plotted in figure 6 as a function of E/N for H₂O, dry air and all gas mixtures. Provided that the range of k_{eff} covers five orders of magnitude, we have used logarithmic scales. The values to the left of the minimum of each curve are negative because $(\alpha-\eta)/N < 0$. It is in this region where one can see the high influence of H₂O in the reaction rate values. Note that the minima on each curve correspond to the limiting field strength, E/N_{lim} , where $\alpha = \eta$, and

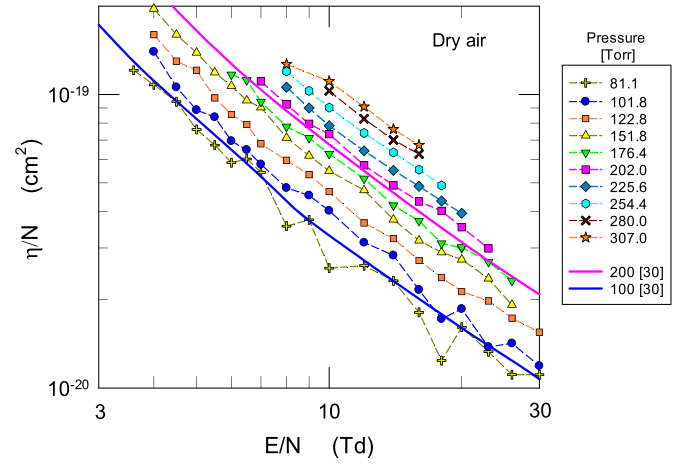


Figure 7. Measured attachment coefficients for dry air as a function of E/N and with gas pressure as a parameter, and the calculations from Taniguchi *et al* [30]. Reproduced from [30]. © IOP Publishing Ltd All rights reserved.

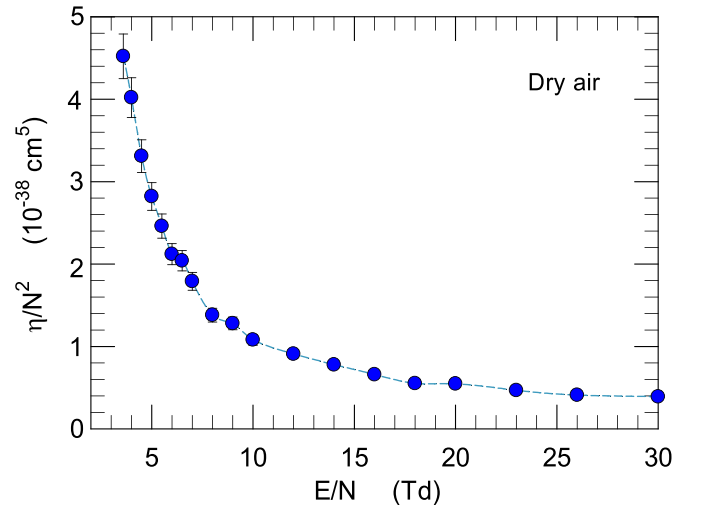


Figure 8. Present values of the density-squared normalised electron attachment coefficients, η/N^2 , for dry air.

that these E/N_{lim} values increase with the amount of H₂O in the mixture, which in turn means that the dielectric strength of the mixtures is higher than that of dry air.

3.3.2. Three-body attachment in dry air. Because of its importance, we will single out the three-body attachment coefficients, η/N , in dry air over the E/N region where there is no ionisation at all, namely, $3 \leq E/N \leq 30$ Td, as is shown in figure 7 over the pressure range 81.1–307 Torr. The present data are compared in the same figure with the calculations of Taniguchi *et al* [30] for 100 and 200 Torr at the same pressures, with which the agreement can be regarded as good, with differences of 9% on average with the 202 Torr measured curve and 12% with the 101.8 Torr one. Dividing the η/N values of figure 7 by their corresponding density N , evidence of three-body attachment in dry air is shown in figure 8, where the

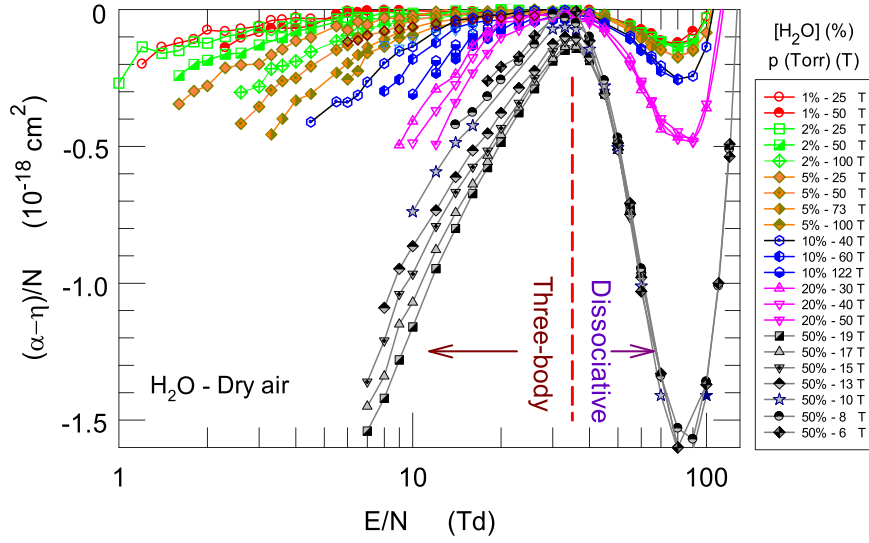


Figure 9. The effective ionisation coefficient, $(\alpha-\eta)/N$, in H_2O -dry air mixtures, shown for the E/N region where $(\alpha-\eta)/N < 0$. The vertical, broken red line indicates the transition region between three-body attachment for $E/N < 30$ Td (left) and two-body (dissociative, right) attachment for $E/N > 35$ Td.

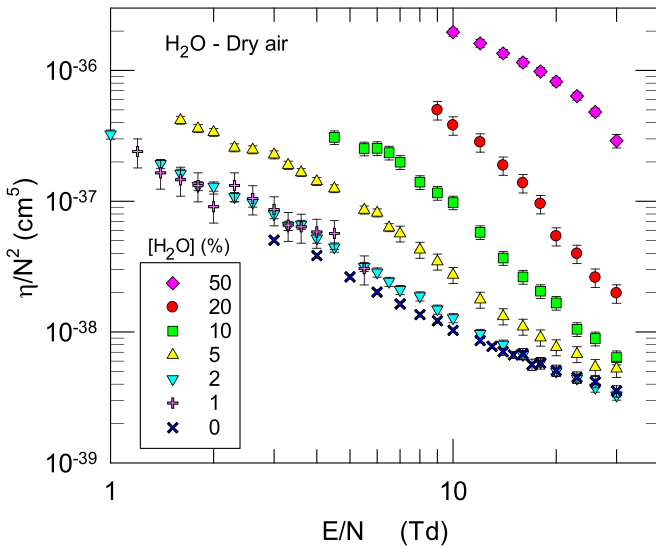


Figure 10. Three-body electron attachment coefficients normalised to the square of gas density for air and the H_2O -air mixtures.

values of η/N^2 are plotted as a function of E/N , resulting in a single curve with uncertainty ranges between 4%–10%.

3.3.3. Three-body attachment in H_2O -dry air mixtures. In figure 9, the E/N region from 1 to 120 Td has been expanded in order to show clearly the dissociative attachment region from about $E/N = 40$ Td onwards, in which the density-normalised attachment coefficient, η/N , does not depend on gas density N (or pressure) since it is due to a binary collision process. On the other hand, for $E/N < 40$ Td, the attachment coefficient bears a strong and clear dependence on N , thereby suggesting

the presence of three-body collisions for which the density-squared normalised attachment coefficient, η/N^2 , should not depend on N . The η/N values for $E/N > 40$ Td have been intentionally plotted without averaging to show that these form a single curve, as expected for a binary collision process, in contrast with those due to three-body attachment, also shown to the left of the plot, for which the pressure dependence is clear.

The η/N attachment coefficients for $E/N < 30$ Td were measured over the pressure range 6–122 Torr and are plotted in figure 9, where one can appreciate the very large effect of water on the η/N values of the mixtures.

Referring again to the $(\alpha-\eta)/N$ coefficients of figure 9, we shall now focus our attention on those belonging to the three-body region, where one clearly sees the dependence on pressure or gas density. Over the E/N range 1–30 Td, ionisation processes are absent ($\alpha = 0$), hence these values can safely be regarded as attachment coefficients, η/N which, after dividing them by N , we obtain the η/N^2 coefficients plotted in figure 10, which is a single curve for each H_2O -dry air combination, thereby confirming the hitherto postulated three-body attachment hypothesis for our measurements. It is noteworthy to see the striking influence of H_2O on the low-energy attachment process with, for instance, a growth in η/N^2 close to or more than two orders of magnitude between dry air and the 50% H_2O -dry air mixture in their overlap range.

The three-body reaction rates, k_{3B} , have been evaluated from

$$k_{3B} = \frac{\eta}{N^2} W_F, \quad (12)$$

and are shown in figure 11.

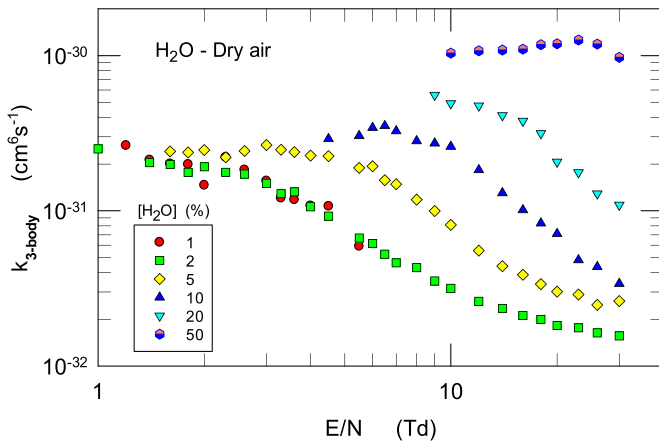


Figure 11. Three-body attachment rates for the H_2O -dry air mixtures.

4. Conclusions

This work has concentrated mainly on the three-body attachment processes leading to the formation of stable O_2^- , both in dry air and its mixtures with H_2O , as well as to dissociative, two-body attachment. Efforts were made to cover as wide E/N and gas pressure ranges as possible. Values of η/N and η/N^2 are presented, together with the corresponding reaction rates. Besides, we have included the newly measured electron drift velocities for the 50% H_2O -dry air mixture along with previously published velocities for the mixtures containing 1%–20% H_2O . The longitudinal diffusion coefficients, derived

from the same set of measurements are herein presented. We know of no other measured ND_L values for the mixtures published in the literature.

Because of the importance of humid air, these mixtures were studied extensively during the years 1960–1980 for only dry air and 1%–2% H_2O . However, no other data for η/N or η/N^2 of mixtures with H_2O abundances higher than 2% have been published before, even though the field of atmospheric physics has grown to embrace the physics of clouds, in which the abundance of water in air is higher than that at ground level, leading to cloud-to-cloud sparks, or even upper atmospheric lightning such as sprites and elves. The present data may also be useful for modelling discharges in the field of bioplasmas and that of plasmas in medicine. Finally, from the point of view of basic physics, these data could be of particular interest in the fields of testing or derivation of cross sections, particularly in the case of H_2O , for which a recent article has stressed the need for more cross section data [31].

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

Acknowledgments

The authors are thankful to A Bustos, G Bustos and H Hinojosa for their technical assistance. This research was supported by Projects DGAPA-UNAM IN 112223, and CONAHCYT, Grant 240073.

Appendix

Table 2. Drift velocity, W , for the H₂O-dry air mixtures.

	0%	1%	2%	5%	10%	20%	50%	100%
E/N (Td)	W (10^5 cm s^{-1})							
1			7.72					
1.2		11.0	9.01					
1.4		12.9	10.6					
1.6		13.8	12.2	5.79				
1.8		15.1	13.5	6.61				
2		16.1	14.8	7.31				
2.3		16.8	16.4	8.61				
2.6		17.5	17.6	9.80				
3	10.9	18.0	18.8	11.7				
3.3	11.2	18.4	19.8	13.1				
3.6	11.8	18.5	20.0	14.4				
4	12.4	18.5	20.4	16.0				
4.5	13.0	18.8	20.8	18.0	9.44			
5	13.6							
5.5	14.3	19.4	21.2	22.0	12.0			
6	15.0	19.5	21.5	23.7	13.5			
6.5	15.6	19.8	21.6	25.0	15.0			
7	16.3	20.5	22.0	26.2	16.4			
8	17.5	21.2	22.9	27.7	20.1	9.57		
9	18.9	22.4	23.7	28.8	23.5	11.2		
10	20.1	23.8	24.7	29.7	26.5	12.9	5.30	
12	22.8	26.1	26.9	31.3	31.7	16.8	6.67	
14	25.3	28.3	29.2	33.3	35.4	21.9	8.05	
16	27.8	31.1	31.6	35.2	38.3	27.5	9.57	3.51
18	30.2	33.3	33.9	37.1	40.5	33.0	12.0	4.04
20	32.6	35.7	36.4	39.4	42.6	38.2	14.5	4.51
23	36.1	39.4	39.9	42.6	45.9	44.5	19.8	5.25
26	39.4	42.8	43.4	45.8	48.8	49.4	24.6	6.06
30	43.7	46.9	47.7	50.0	53.1	55.1	33.6	7.15
33	46.8	50.3	51.3	53.4	56.2	58.8	40.7	8.14
36	50.1	54.0	54.8	56.5	59.1	62.4	46.6	9.19
40	54.1	58.2	58.9	60.7	63.4	66.8	56.2	10.8
45	59.1	63.4	64.3	65.8	68.5	71.8	68.0	14.1
50	63.8							19.3
55	68.6	73.3	74.6	75.6	78.1	82.0	82.2	27.3
60	73.4	79.1	79.5	81.0	83.1	87.1	89.0	39.1
65	78.1	84.2	84.8	85.8	87.9	92.1	95.0	51.7
70	83.3	88.8	89.6	91.7	92.8	96.8	101	67.4
80	93.2	100	100	102	103	108	112	92.1
90	103	111	111	113	114	117	124	113
100	114	122	123	123	125	129	132	128
120	135	144	144	145	146	150	152	155
130								172
140	155	163		163	169	172	173	181
150								193
160	175						185	204
180	195						198	223
200	215						217	242
230	244						241	276
260	271						273	305
300	305						296	341
330	336							373
360	359							407
400	393							430
450	441							484
500								520
550								571
600								608
650								654

Table 3. Longitudinal diffusion coefficient, ND_L , for the H₂O-dry air mixtures.

	0%	1%	2%	5%	10%	20%	50%	100%
E/N (Td)	ND_L ($10^{21} \text{ cm}^{-1} \text{ s}^{-1}$)							
1.8		4.81						
2		4.81						
2.3		4.02						
2.6		4.93						
3		4.07						
3.3		5.26	3.64					
3.6		5.26	3.64					
4		5.26	5.49					
4.5		3.92	5.49	4.82				
5.5		5.62	5.49	4.82	1.03			
6	8.27	5.62	5.49	5.84	1.63			
6.5	8.10	6.56	5.49	4.80				
7	0.00	6.56	7.47	5.60	3.21			
8	7.11	8.47	7.47	5.60	5.16			
9	7.65	6.64	7.47	5.99	5.20	0.73		
10		6.64	7.47	5.99	5.42	1.00		
12	8.29	8.28	8.82	5.99	6.12	1.67		
14	7.64	8.28	8.10	6.98	5.89	2.45		
16	7.80	8.28	8.10	6.98	6.47	4.31		
18	7.67	8.28	8.10	6.46	6.79	6.47		
20	7.71	8.28	8.10	6.46	7.50	8.16	1.70	
23	6.15	6.81	8.10	8.79	8.55	8.48	2.55	
26	5.51	6.81	8.10	8.79	8.81	8.74	3.70	0.31
30	6.92	6.81	8.10	8.79	8.80	8.81	6.00	0.56
33	5.79	6.81	8.10	8.79	8.88	8.60	10.8	0.85
36		6.81	8.10	8.79	8.33	8.87	12.8	1.05
40	6.24	6.81	9.64	9.47	8.42	9.35	14.4	1.97
45	9.23	6.81	9.64	8.59	8.61	8.35	18.0	4.21
50	8.41							9.38
55	8.39	10.3	13.1	10.2	10.8	10.6	17.5	19.5
60	10.6	11.4	13.1	10.2	12.9	11.3	18.7	33.0
65		10.3	12.4	10.2	14.0	12.2		37.8
70		11.4	9.71	13.1	12.2	13.9	20.4	39.1
80	13.0	14.0	17.1	16.4	13.1	17.0	23.3	41.2
90		20.1	17.1	20.2	18.5	24.0	25.3	39.8
100	18.4	21.2	28.5	21.5	25.9	25.4	27.6	40.7
120	18.5			25.4	27.6	31.8	28.6	39.3
130								43.4
140	19.6				32.4		32.3	41.1
150								38.9
160	23.5						34.0	42.3
180	25.6							42.9
200	31.4							44.2
230	32.0							47.9
260	34.2							49.5
300	37.7							51.0
330	42.6							52.9
360	41.7							57.5
400	45.6							54.7
450	46.8							61.0
500								60.7
550								63.5
600								66.1
650								65.1

Table 4. Effective ionisation coefficients, $(\alpha-\eta)/N$, for the H₂O-dry air mixtures.

E/N (Td)	0%	1%	2%	5%	10%	20%	50%	100%
	$\frac{(\alpha-\eta)}{N}$		(10^{-19} cm^2)					
16								−11.6
18								−9.57
20								−9.24
23								−9.63
26								−8.35
30								−6.86
33								−6.19
36								−5.83
40		−0.03	−0.16	−0.18	−0.24	−0.41		−5.86
45		−0.25	−0.26	−0.29	−0.53	−0.80		−7.73
50								−14.2
55		−0.54	−0.61	−0.72	−1.01	−1.94	−2.77	−19.6
60		−0.74	−0.85	−1.01	−1.49	−2.64	−3.33	
65		−0.93	−1.01	−1.28	−1.72	−3.36		−25.7
70		−1.18	−1.18	−1.48	−2.10	−4.10	−3.27	
80		−1.24	−1.39	−1.70	−2.50	−4.74	−9.14	−35.1
90		−0.94	−1.19	−1.49	−2.31	−4.79	−9.63	−40.9
100		−0.27	−0.31	−0.48	−1.44	−3.61	−9.42	−37.3
120	1.81	3.32	3.44	3.30	3.18	2.03	−2.87	−20.1
130								−5.29
140	6.63	10.3	10.4	11.4	11.2	12.1	7.9	8.16
150								25.9
160	14.7						21.9	42.0
180	26.5						31.3	74.9
200	38.6						62.3	113
230	60.8						95.8	179
260	92.7						156	234
300	138						213	320
330	175							379
360	196							447
400	258							503
450	328							604
500								703
550								835
600								948
650								1070

Table 5. Three-body attachment coefficients, η/N^2 for the H₂O-dry air mixtures.

E/N (Td)	0%	1%	2%	5%	10%	20%	50%
				$\frac{\eta}{N^2}$	(10^{-39} cm^5)		
1			325				
1.2		240					
1.4		165	193				
1.6		146	163	417			
1.8		132	131	360			
2		90.9	130	337			
2.3		132	108	257			
2.6		105	98.0	248			
3	50.30	86.6	79.9	227			
3.3		65.7	65.1	189			
3.6		63.8	66.3	166			
4	38.4	58.3	52.0	142			
4.5		56.9	44.4	125	308		
5	26.4						
5.5		30.5	31.5	86.0	253		
6	20.2		28.6	81.7	254		
6.5			24.3	63.0	235		
7	16.4		21.0	56.7	199		
8	13.6		18.8	42.6	140		
9	12.2		14.9	34.6	116	498	
10	10.3		12.8	27.3	97.8	381	1960
12	8.62		9.70	17.7	57.9	283	1610
13	7.78						
14	7.07		8.04	13.2	36.8	188	1350
15	6.69						
16	6.81		6.69	11.0	26.4	138	1150
17	5.64						
18	5.79		5.89	9.08	20.5	95.6	981
20	5.08		5.01	7.65	16.7	54.1	822
23	4.51		4.42	6.78	10.5	39.7	637
26	4.16		3.77	5.40	8.91	26.1	482
30	3.54		3.28	5.24	6.39	19.8	290

ORCID iDs

J de Urquijo  <https://orcid.org/0000-0003-3379-9843>

O González-Magaña  <https://orcid.org/0000-0001-6722-2327>

E Basurto  <https://orcid.org/0000-0002-2972-5545>

A M Juárez  <https://orcid.org/0000-0003-1677-4935>

References

- [1] Von Woedtke T, Laroussi M and Gherardi M 2022 *Plasma Sources Sci. Technol.* **31** 054002
- [2] Yusupov M, Neyts E C, Simon P, Berdiyov G, Snoeckx R, van Duin A C T and Bogaerts A 2014 *J. Phys. D: Appl. Phys.* **47** 025205
- [3] Ferreira C M, Gordiets B, Tatarova E, Henriques J and Días F M 2002 *Chem. Phys.* **398** 248
- [4] Yang Y, Cho Y I and Fridman A 2017 *Plasma Discharge in Liquid: Water Treatment and Applications* (Taylor and Francis)
- [5] Aleksandrov N L, Bazelyan E M, Ponomarev A A and Starikovskiy A Y 2022 *J. Phys. D: Appl. Phys.* **55** 383002
- [6] Gordillo-Vázquez F J 2008 *J. Phys. D: Appl. Phys.* **41** 234016
- [7] Kuffel E 1959 *Proc. Phys. Soc.* **74** 297
- [8] Chen X, He W, Du X, Yuan X, Lan L, Wen X and Wan B 2018 *Phys. Plasmas* **25** 063525
- [9] Babichev V N, Galeeva K E, Kirichenko A N, Nekrasov A A, Ugodchikova A V, Trushkin N I, Filippov A V, Cherepanova Y V and Cherkovets V E 2023 *Plasma Phys. Rep.* **49** 563
- [10] Zhou R, Wang P, Xian Y, Mai-Prochnow A, Lu X, Cullen P J, Ostrikov K and Bazaka K 2020 *J. Phys. D: Appl. Phys.* **53** 303001
- [11] Ali M, Cheng J-H, Tazeddinova D, Aadil R M, Zeng X-A, Goksen G, Lorenzo J M, Johnson Esua O and Manzoor M F 2023 *Ultrason. Sonochem.* **97** 106461
- [12] Dutton J 1975 *J. Phys. Chem. Ref. Data* **4** 517–856
- [13] Gallagher J W, Beaty E C, Dutton J and Pitchford L C 1983 *J. Phys. Chem. Ref. Data* **12** 109–51
- [14] Itikawa Y and Mason N 2005 *J. Phys. Chem. Ref. Data* **34** 1–22
- [15] Chanin L M, Phelps A V and Biondi M A 1959 *Phys. Rev. Lett.* **2** 344–6
- [16] Chanin L M, Phelps A V and Biondi M A 1962 *Phys. Rev.* **162** 219–30
- [17] Spence D and Schulz G J 1972 *Phys. Rev. A* **5** 724–32

- [18] Grünberg R 1969 *Z. Naturforsch.* **24a** 1039–48
- [19] Taniguchi T, Tagashira H, Okada I and Sakai Y 1978 *J. Phys. D: Appl. Phys.* **11** 2281
- [20] Davies D K 1983 Theoretical Notes Note 346 (Westinghouse Electric Corp)
- [21] Melton C E 1972 *J. Chem. Phys.* **57** 4218–25
- [22] Compton R N and Christophorou L G 1967 *Phys. Rev.* **154** 110
- [23] Hagelaar G J M and Pitchford L C 2005 *Plasma Sources Sci. Technol.* **14** 722
- [24] Jafke T, Meinke M, Hashemi R, Christophorou L G and Illenberger E 1992 *Chem. Phys. Lett.* **193** 62–68
- [25] Hernández-Ávila J L, Basurto E and de Urquijo J 2002 *J. Phys. D: Appl. Phys.* **35** 2264–9
- [26] Brambring J 1964 *Z. Phys.* **179** 532–8
- [27] Raether H 1964 *Electron Avalanches and Breakdown in Gases* (Butterwords)
- [28] Christophorou L G 1984 *Electron-Molecule Interactions and Their Applications* vol 2 (Academic Press) pp 90–207
- [29] Ruiz-Vargas G, Yousfi M and de Urquijo J 2010 *J. Phys. D: Appl. Phys.* **43** 455201
- [30] Taniguchi T, Kawamura K, Sakamoto S and Tagashira H 1982 *J. Phys. D: Appl. Phys.* **15** 1187–93
- [31] Budde M, Dias T C, Vialetto L, Pinhão N, Guerra V and Silva T 2022 *J. Phys. D: Appl. Phys.* **55** 445205